Sidewalk number stories

How a Sidewalk Is Made and How It Works

Name _____ Date _____

Use numbers or a drawing to share your thinking.

1. Little counts 36 tiny stones on one chip of concrete and 29 on another chip. How many stones is that in all?

36 + 29 = _____

Show your thinking.

2. A long sidewalk has 82 squares, and 47 of them have been swept. How many squares are left to sweep?

82 − 47 = _____

Show your thinking.

3. Big puts 45 pebbles in one jar and Little puts 38 pebbles in another jar. How many pebbles are in both jars?

45 + 38 = _____

Show your thinking.

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4. A sample tray holds 71 pebbles. Little sorts 28 of them into a jar. How many pebbles are left in the tray?

71 − 28 = _____

Show your thinking.

5. A new slab is kept damp for 7 days. It then has 21 more days to keep getting stronger. How many days is that in all?

7 + 21 = _____

Show your thinking.

6. A new path needs 72 grooves. Workers have pressed 48 so far. How many grooves are left to press?

72 − 48 = _____

Show your thinking.

Four ingredients

A sidewalk is concrete: gravel, sand, gray cement powder and water mixed together. Cement is only one ingredient, not the whole thing.

Paste that glues

Cement and water make a gray paste. It coats every stone and grain of sand and, once hard, holds them all tightly together.

Pour, spread, wait

Wet concrete is poured into a wooden form and smoothed. Cement and water then react and link together, so it hardens, even under water.

Curing takes days

A new slab is kept damp for about a week while the reaction goes on. This is curing. Most strength builds in the first month.

A line for cracks

Concrete shrinks as it hardens and can crack. Grooves make thin lines in the slab, so cracks tend to follow them in neat, straight lines.

Strong when squeezed

Concrete is strong when squeezed but weaker when bent. Packed gravel under the slab helps spread the weight so it bends less.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

How a Sidewalk Is Made and How It Works

Explain a sidewalk in your words

How a Sidewalk Is Made and How It Works · Curiosity day

Name _____ Date _____

Choose one clue. Say your idea first, then draw or write it.



Paste that glues



Poured into a frame



A line for cracks

Draw one square of sidewalk and add a small arrow to the groove.

concrete · cement · form · chemical reaction · curing · groove · because · so

Cement and water make a paste that … what does the paste hold?

After it is poured, the concrete … how does it turn hard?

A groove helps because … where does a crack go?

Does your sentence tell what happens and use because or so to explain why?

How a Sidewalk Is Made and How It Works

01 What is a sidewalk made of?



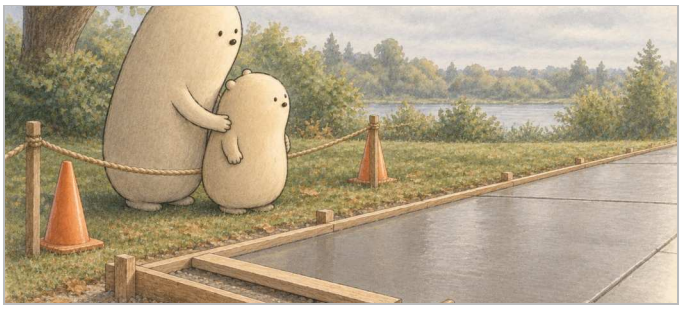
Concrete is gravel, sand, cement and water.

Look closer. Find the tiny stones and sandy specks on the chipped corner, then look at the four sealed jars beside it.

A little explanation. A sidewalk is made of concrete, and concrete is a mix of four things: gravel, sand, cement and water. Cement is a fine gray powder. It is only a small part of the mix, roughly one part in eight, and when it meets water it makes a paste that coats the sand and stones.

Try this. Look closely at a sidewalk square outside, or at a photo of one. Find tiny stones, speckles and a groove. Draw one square on paper and mark where the groove runs.

02 How does wet concrete turn hard?



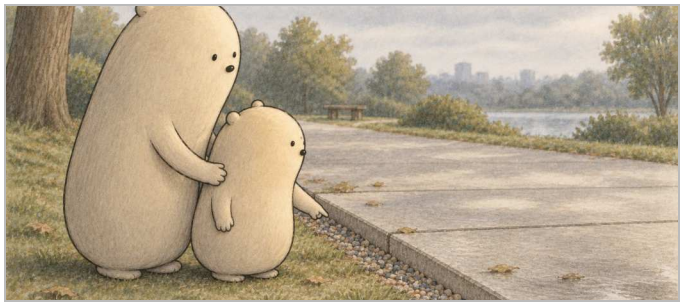
Wet concrete fills a wooden form and then hardens by a chemical reaction.

Look closer. Find the smooth wet slab inside its wooden frame, and the rope and cones that keep everyone back.

A little explanation. Wet concrete is poured into a wooden frame called a form, spread flat and smoothed. Then the cement and water begin a chemical reaction that links everything together. It hardens by that reaction, not just by drying out, which is why concrete can even harden under water.

Try this. Stir two spoonfuls of flour with a little water in a paper cup to make a thick paste. Stir in a handful of dry rice so the paste coats every grain. Leave the cup for a day, then tap it and feel how hard it got. Flour paste only dries out, but real concrete hardens by a chemical reaction.

03 Why is a sidewalk strong, and what holds it up?



Concrete is strong when squeezed, and packed gravel helps support the slab evenly.

Look closer. Look at the edge of the path, where a thin layer of packed gravel sits under the slab.

A little explanation. Concrete is very strong when it is squeezed, like when feet and bike wheels press down on it. It is weaker when it bends or stretches. A layer of packed gravel under the slab helps support it evenly and spread the weight out, so the slab bends less.

Try this. Stand a stick of sidewalk chalk on end and press down on top with one finger to see that it holds. Now lay the chalk across two books and press gently in the middle until it snaps. Chalk is a stand-in for concrete: strong when squeezed, weak when bent.

04 Why does a sidewalk have grooves?



A groove makes a planned weak line, so a crack tends to follow it.

Look closer. Follow the straight groove between two squares and find the thin crack that stays inside it.

A little explanation. A new slab shrinks a little as it hardens, and heat and cold make it grow and shrink all year. Concrete can crack when that happens. A groove makes the slab thinner along one line, so a crack tends to start there and run straight instead of wandering anywhere.

Try this. Look at the lines on a graham cracker. Break one cracker along a line. Then break another across the middle where there is no line. See which break is neat and straight.